Experimental Examples and Demonstrations

1. Measurement of Refractive Index

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### Example box: Abbe Refractometer
The Abbe refractometer provides a precise method for measuring refractive indices of liquids

1. Operating principle: Total internal reflection at critical angle
2. Typical accuracy: ±0.0002
3. Procedure:
   - Place sample between prisms
   - Adjust viewing angle until bright/dark boundary appears
   - Read refractive index directly from scale
4. Applications:
   - Quality control in food/chemical industries
   - Concentration measurements
   - Purity testing
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2. Dispersion Demonstration

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### Example box: Prism Spectroscopy
Simple but effective demonstration using:
- White LED source
- Glass prism
- Screen

Observations:
1. Separation of white light into spectrum
2. Different angles of deviation for different colors
3. Measurement of dispersion relation n( )
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3. Absorption Measurements

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### Example box: Beer-Lambert Law Verification
Equipment:
- Spectrophotometer
- Solutions of varying concentration
- Cuvettes of different path lengths
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Procedure:
1. Measure transmission vs. concentration
2. Plot ln(I/I ) vs. path length
3. Determine absorption coefficient
4. Verify linear relationship
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4. Group Velocity Measurement

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### Example box: Fiber Optic Pulse Propagation
Setup:
- Short pulse laser source (~ps)
- Optical fiber of known length
- Fast photodetector
- Oscilloscope

Measurements:
1. Pulse arrival time vs. fiber length
2. Pulse width changes due to dispersion
3. Calculate group velocity
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5. Practical Applications

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### Example box: Optical Communications
Demonstration of dispersion effects in telecommunications:
1. Signal transmission through optical fiber
2. Pulse broadening measurement
3. Dispersion compensation techniques
4. Bit error rate measurements
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6. Advanced Demonstrations

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### Example box: Anomalous Dispersion
Using atomic vapor cells:
1. Sodium vapor cell setup
2. Tunable laser source
3. Measurement of refractive index near D-lines
4. Observation of rapid n( ) variation
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You might also add hands-on activities:

1. **Student Exercises**:
   - Measuring refractive indices of different materials
   - Plotting dispersion curves
   - Analyzing absorption spectra

2. **Virtual Labs**:
   - Interactive simulations of wave propagation
   - Online spectroscopy experiments
   - Data analysis exercises

3. **Real-world Connections**:
   - Fiber optic communications
   - Spectroscopy in chemistry
   - Optical imaging systems
   - Medical diagnostics

4. **Troubleshooting Scenarios**:
   - Common experimental problems
   - Error analysis
   - Calibration procedures

These examples would help students connect theoretical concepts with practical applications and develop experimental skills.